

Scalar SRF: $S = \int d^d x \frac{1}{2} (\partial\phi)^2$

E.O.M.: $\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0$

$\partial_\mu (\partial^\mu \phi) = 0 \Rightarrow \partial^2 \phi = 0 \quad T_\mu = T_{\mu\nu}^\mu$

Coupled to matter: $S = \frac{1}{c} \int d^d x \left\{ \frac{1}{2\alpha c^2} (\partial\phi)^2 + \frac{\phi}{c^2} T_{\text{matter}} + \mathcal{L}_{\text{matter}} \right\}$

E.O.M. $\frac{1}{\alpha c^2} \partial^2 \phi = \frac{1}{c^2} T_{\text{matter}} \Rightarrow \partial^2 \phi = \alpha T_{\text{matter}}$

• Regard ϕ to be the Newtonian potential to find α

$(T_{\text{matter}})_{00} = +\rho c^2 \Rightarrow T_\mu^\mu = +\rho c^2$

$\partial^2 \phi = +\rho c^2 \alpha$

for a static Newtonian potential

$\partial_i \partial_i \phi = \rho c^2 \alpha$

$\partial_1^2 \phi + \dots + \partial_d^2 \phi = \rho c^2 \alpha$

$\nabla^2 \phi = 4\pi G \rho$ (Poisson eq)

So in $d=4$ $\alpha = 4\pi G_N / c^2$

$\vec{F} = \frac{d\vec{p}}{dt} \quad \rho = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

so $\frac{\partial}{\partial \phi} \left(\frac{1}{2} (-\dot{\phi}^2) \right) + \dots = -\dot{\phi}$
 $\Rightarrow F = \ddot{\phi} = 0?$

Symmetries & Noether: $S[\phi] = \int_{\Sigma} d^d x \mathcal{L}(\phi, \partial\phi, \partial^2\phi, \dots)$

$\delta S = \int_{\Sigma} d^d x \delta \mathcal{L} = \int_{\Sigma} d^d x \left\{ \frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \delta (\partial_\mu \phi) + \dots \right\}$

no variation of coordinates are zero by hypothesis which

means $\delta \partial_\mu = \partial_\mu \delta$

$\delta S = \int_{\Sigma} d^d x \left\{ \left(\frac{\delta \mathcal{L}}{\delta \phi} - \partial_\mu \left(\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \right) \right) \delta \phi + \partial_\mu \left[\frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \delta \phi + \dots \right] \right\}$

$\hat{\delta} x^\mu = x'^\mu - x^\mu \quad | \quad \hat{\delta} \phi(x) = \phi'(x) - \phi(x)$

Poincaré: $x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad \Lambda \eta \Lambda = \eta$

$\hat{\delta} x^\mu = \Lambda^\mu_\nu x^\nu + a^\mu - x^\mu$

$\Lambda^\mu_\nu = (e^{+\sigma \tau})^\mu_\nu = (\mathbb{1} + \sigma \epsilon)^\mu_\nu + \mathcal{O}(\epsilon^2)$

$\hat{\delta} x^\mu = \sigma^\mu_\nu x^\nu + \sigma^\mu, \quad \sigma^{\mu\nu} = -\sigma^{\nu\mu}$

$\tilde{\delta}(d^d x) = \tilde{\delta}(dx^1 \wedge \dots \wedge dx^d) = \tilde{\delta}(dx^1) \wedge \dots \wedge dx^d + \dots$

$\tilde{\delta}(dx^i) = d\tilde{x}^i = d(\sigma^i_\nu x^\nu + \sigma^i)$

$$= \sigma_{\nu}^i dx^{\nu}$$

$$\Rightarrow \sigma_{\nu}^i dx^{\nu} \wedge dx^2 \wedge \dots \wedge dx^d + \dots$$

$$= 0 \quad \nu \neq i$$

$$\therefore \sigma_{\nu}^i dx^1 \wedge dx^2 \wedge \dots \quad | \quad \sigma_{\nu}^i = \eta_{i\nu} \sigma^{\nu} = 0$$

$$\text{and so } \hat{S}(d^d x) = 0 \Rightarrow \hat{S}S = 0$$

$$\bullet \hat{S}\phi = 0 \quad \text{since } \phi \text{ is a scalar and } \phi' = \phi \quad \forall x$$

$$j^{\mu}_{\text{can}} = T^{\mu}_{\text{can}} \tilde{S}_{\nu} x^{\nu} = T^{\mu}_{\text{can}} x^{\nu}$$

$$\text{Full calculation: } \hat{S}S = \int_{\Sigma} d^d x \partial_{\mu} S^{\mu}(\tilde{S})$$

$$* \hat{S}S = \int_{\Sigma} (\hat{S}(d^d x) \mathcal{L} + d^d x \hat{S}\mathcal{L})$$

$$= \int_{\Sigma} d^d x (\partial_{\mu}(\tilde{S} x^{\mu} \mathcal{L}) + \mathcal{L} + \tilde{S} x^{\mu} \partial_{\mu} \mathcal{L})$$

Now we can expand $\mathcal{L}$ at a singular pt. For

$$\mathcal{L} = \mathcal{L}(\phi, \partial_{\mu} \phi, \partial_{\mu} \partial_{\nu} \phi, \dots)$$

$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \delta(\partial_{\mu} \phi) + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \delta(\partial_{\mu} \partial_{\nu} \phi) + \dots$$

$$= \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \partial_{\mu} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\mu} \partial_{\nu} (\delta \phi) + \dots$$

$\tilde{S}$: "total" variation, usually only deal with "one"

$$\hat{S}S = \int_{\Sigma} d^d x \left(\partial_{\mu} (\tilde{S} x^{\mu} \mathcal{L}) + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \partial_{\mu} \delta \phi + \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\mu} \partial_{\nu} \delta \phi + \dots \right)$$

$$= \int_{\Sigma} d^d x \left(\partial_{\mu} (\tilde{S} x^{\mu} \mathcal{L} + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \delta \phi + \dots) + \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi - \partial_{\mu} \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \delta \phi \right) + \dots$$

$$\text{Second order term: } \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\mu} \partial_{\nu} \tilde{S} \phi = \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\nu} \tilde{S} \phi \right) - \partial_{\mu} \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\nu} \tilde{S} \phi$$

$$= \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\nu} \delta \phi \right) - \partial_{\nu} \left(\partial_{\mu} \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \delta \phi \right) + \partial_{\nu} \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \right) \delta \phi$$

$$\hat{S}S = \int_{\Sigma} d^d x \left\{ \partial_{\mu} \left[\tilde{S} x^{\mu} \mathcal{L} + \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} - \partial_{\nu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \right) \right) \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \partial_{\nu} \delta \phi + \dots \right] \right.$$

$$\left. + \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi - \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \right) \delta \phi + \partial_{\mu} \partial_{\nu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \right) \delta \phi + \dots \right\}$$

$$\text{Define: } \frac{\delta S}{\delta \phi} := \frac{\partial \mathcal{L}}{\partial \phi} - \partial_{\mu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \phi)} \right) + \partial_{\mu} \partial_{\nu} \left(\frac{\partial \mathcal{L}}{\partial (\partial_{\mu} \partial_{\nu} \phi)} \right) - \dots$$

$$\frac{\delta S}{\delta \phi} = \sum_{i=0}^{\infty} (-1)^i \partial_{\mu_i} \frac{\partial \mathcal{L}}{\partial (\partial_{\mu_i} \phi)}$$

$$\hat{S}S = \int_{\Sigma} d^d x \left(\partial_{\mu} T^{\mu}_{\text{can}}(\tilde{S}) + \frac{\delta S}{\delta \phi} \delta \phi \right) = 0$$

with

$$\Gamma_N^{\mu}(\tilde{\phi}) = -\tilde{S}^{\mu}(\tilde{\phi}) + (T_{\text{can}})^{\mu}_{\nu} \tilde{S}^{\nu} - \frac{\partial \tilde{S}}{\partial \partial_{\mu} \tilde{\phi}} \partial_{\mu} \tilde{S}^{\nu} + \left(\frac{\partial \tilde{S}}{\partial \partial_{\mu} \tilde{\phi}} - \partial_{\nu} \left(\frac{\partial \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \tilde{\phi}} \right) \right) \tilde{S}^{\nu} + \dots$$

where, for all terms within the Boundary term we have used

$$\tilde{S} = S + \tilde{S}^{\mu} \partial_{\mu}$$

$$(T_{\text{can}})^{\mu}_{\nu} = M^{\mu}_{\nu} \tilde{S} - \frac{\partial \tilde{S}}{\partial \partial_{\mu} \tilde{\phi}} \partial_{\nu} \tilde{\phi} - \frac{\partial \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \tilde{\phi}} \partial_{\nu} \partial_{\mu} \tilde{\phi} + \partial_{\rho} \left(\frac{\partial \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}} \right) \partial_{\nu} \tilde{\phi} + \dots$$

On-shell: $\frac{\delta S}{\delta \phi} = 0$

$j_N^{\mu}(\tilde{\phi})$: conserved vector current $\left| \partial_{\mu} j_N^{\mu}(\tilde{\phi}) = 0 \right.$

Global transformations:

$$\tilde{S}^{\mu} x^{\nu} \equiv \sigma^{\mu}{}_{\alpha} \tilde{S}^{\alpha} x^{\nu}$$

$$\tilde{S}^{\mu} \phi \equiv \sigma^{\mu}{}_{\alpha} \tilde{S}^{\alpha} \phi$$

$$* T^{\mu}_{\nu} = S^{\mu}_{\nu} \tilde{S} - \frac{\partial \tilde{S}}{\partial \partial_{\mu} \tilde{\phi}} \partial_{\nu} \tilde{\phi} - \frac{\partial \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \tilde{\phi}} \partial_{\nu} \partial_{\mu} \tilde{\phi}$$

E.O.M $\partial^2 \phi = 0$

$$T_{\mu\nu} = M_{\mu\nu} \left(\frac{1}{2} (\partial \phi)^2 \right) - \partial_{\mu} \phi \partial_{\nu} \phi$$

$$\partial^{\mu} T_{\mu\nu} = \frac{1}{2} \partial_{\nu} (\partial \phi)^2 - \partial^2 \phi \partial_{\nu} \phi - \partial^{\mu} \phi \partial_{\mu} \partial_{\nu} \phi$$

$$= \partial_{\nu} \partial_{\mu} \phi \partial^{\mu} \phi - \partial^2 \phi \partial_{\nu} \phi - \partial_{\mu} \partial_{\nu} \phi \partial^{\mu} \phi = 0$$

on-shell

$$Q_{(n)} = \int_{V_t} d^{d-1} x j_{(n)}^0 = \int_{V_t} d^{d-1} x T_{\text{can}}^0{}_{\nu}$$

$$\sim \partial_i \partial_i \phi = \alpha c^2 \rho$$

$$(\sum_i p_i^2) \phi = \alpha c^2 K(\rho)$$

$$\phi(x) = \tilde{\alpha} \int d^d p \frac{e^{i p \cdot x}}{\sum_i p_i^2} \int \frac{d^d y}{(2\pi)^d} e^{-i p y} \rho(y)$$

$$= \tilde{\alpha} \int \frac{d^d y}{(2\pi)^d} \rho(y) \int d^d p e^{-i p(y-x)} \frac{1}{p_1^2 + \dots + p_d^2}$$

generalized spherical

$$= \frac{2\tilde{\alpha}}{(2\pi)^d} \int d^d y \rho(y) \int d^{d+1} p \int_0^{\infty} d\omega |\mathbf{p}|^{d-3} e^{-i|\mathbf{p}|(\omega - (x-y) \cdot \hat{\mathbf{p}})} |\mathbf{p}|^{d-1} \sin \dots d|\mathbf{p}| d\theta \dots$$

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

$$z \int_0^{\infty} \alpha^{z-1} t^{z-1} e^{-\alpha t} \alpha dt$$

$$\alpha^z \int_0^{\infty} t^{z-1} e^{-\alpha t} dt$$

$$z(z-2) = (1) \frac{\Gamma(-z) \Gamma(1+z)}{\Gamma(3-z)}$$

$$\int d^d x \frac{\partial^2 \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}} \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi} \sim \int d^d x \frac{\partial^2 \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}} \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}$$

$$\int d^d x \frac{\partial^2 \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}} \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi} \sim \int d^d x \frac{\partial^2 \tilde{S}}{\partial \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}} \partial_{\mu} \partial_{\nu} \partial_{\rho} \tilde{\phi}$$

$$\int \frac{d^d p}{(2\pi)^d} \frac{e^{-p \cdot (x-y)}}{(p^2 + \dots)^{\frac{d-3}{2}}} \quad \int d\vec{p} \frac{e^{-\vec{p} \cdot \vec{x}}}{(p_1^2 + \dots)}$$

Point particle with $p = M \delta^{(d-1)}(\vec{x}_{d-1})$

$$\phi = \frac{16\pi G_N^{(d)} M}{2(d-2) \omega_{d-2}} \frac{1}{|\vec{x}_{d-1}|^{d-3}}$$

$$\omega_{(d)} = 4\pi$$

$$d=4: \quad \phi = \frac{4\pi G_N^{(4)} M}{4\pi |\vec{x}|} = \frac{G_N^{(4)} M}{|\vec{x}|}$$

which is Newtonian

Action of a point particle: $S[X^\mu(\xi)] = -Mc \int d\xi \sqrt{\eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu}$ $\dot{X}^\mu \equiv \frac{dX^\mu}{d\xi}$

$$P_\mu = \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu}: \quad P_\mu = -Mc \frac{1}{\sqrt{\eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu}} 2 \eta_{\mu\nu} \dot{X}^\nu$$

$$P_\mu = -Mc \frac{\eta_{\mu\nu} \dot{X}^\nu}{\sqrt{\eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu}}$$

→ Note there is no force on this particle from the E.O.M:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial X^\mu} = 0 \quad | \quad \partial_\xi \left(\frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} \right) = 0$$

$$\text{or } \frac{dP_\mu}{d\xi} = 0 \quad (\text{E.O.M})$$

Conservation of ang. momenta: $\tilde{S}_{(\rho\sigma)} X^\mu = \int dV (M_{\rho\sigma})^\mu{}_\nu X^\nu$

$$\int dV (M_{\rho\sigma})^\mu{}_\nu = 2 \tilde{S}_{[\rho}^\mu \tilde{S}_{\sigma]}^\nu$$

$$\text{and so } \dot{\psi}_\alpha^\mu = (\tilde{T}_{\alpha\mu})^\mu{}_\nu \tilde{S} X^\nu = (\tilde{T}_{\alpha\mu})^\mu{}_\nu (2 \tilde{S}_{[\rho}^\nu \tilde{S}_{\sigma]}^\lambda) X^\lambda$$

$$(M_{\rho\alpha}) = 2 P_\nu \tilde{S}_{[\rho}^\nu \tilde{S}_{\alpha]}^\lambda X^\lambda = 2 P_\rho X_{\alpha\lambda}$$

$$M_{\mu\nu} = 2 P_{[\mu} X_{\nu]} \quad d(d+1)/2 \text{ components of ang. momenta.}$$

$$\delta X^\mu = \sigma^\mu{}_\nu X^\nu + \sigma^\mu, \quad \sigma^{\mu\nu} = -\sigma^{\nu\mu}$$

$$J(\sigma)/d\xi = 0$$

$$J(\sigma) = P_\mu \delta X^\mu$$

Translations: $J(\sigma^\mu) \sim P^\mu$

Lorentz: $J(\sigma^{\mu\nu}) \sim M^{\mu\nu}$

* $\mathcal{M}$ has $d(d+1)/2$ independent isometries that generate the d -dim poincare group $\mathcal{M}(w(x), w(y)) = \mathcal{M}(x, y)$

diffco/reparametrization invariance: $\xi'(\xi)$

→ local gauge transformations

$$\delta \xi = \epsilon(\xi)$$

* X are scalars under this transformation i.e. $\frac{1}{|\tilde{S}|}$

$$X(\xi')^\mu = X(\xi)^\mu$$

$$|\tilde{S}| =$$

$$\delta F = F'(x') - F(x) \quad | \quad \delta F = F'(\xi) - F(\xi)$$

$$\delta X^\mu = X'^\mu(\xi') - X^\mu(\xi) = 0 = \delta X^\mu$$

$$\delta \dot{X}^\mu = \frac{dX^\mu}{d\xi}(\xi') - \frac{dX^\mu}{d\xi}(\xi) \quad | \quad \xi'(\xi) - \xi = \delta \xi = \epsilon(\xi)$$

$$\delta \dot{X}^\mu = 0$$

$$X'^\mu(\xi') = X^\mu(\xi)$$

$$\delta \dot{X}^\mu = \dot{X}'^\mu(\xi) - \dot{X}^\mu(\xi)$$

$$= \dot{X}^\mu(\xi - \epsilon) - \dot{X}^\mu(\xi) = \dot{X}^\mu(\xi) - \epsilon \ddot{X}^\mu - \dot{X}^\mu$$

$$= -\epsilon \ddot{X}^\mu$$

and $\delta X^\mu = -\epsilon \dot{X}^\mu$

* Invariance of the action means that:

$$\delta S = -Mc \int d\xi \frac{1}{\sqrt{1}} \eta_{\mu\nu} \dot{X}^\mu \delta \dot{X}^\nu$$

$$= -Mc \int d\xi \left(\frac{1}{\sqrt{1}} \right) \eta_{\mu\nu} \dot{X}^\mu (-\epsilon \ddot{X}^\nu) = Mc \int d\xi \sqrt{1} \dot{X}^\mu \ddot{X}^\mu$$

$$= Mc \int d\xi \frac{d}{d\xi} \left(\sqrt{1} \dot{X}^\mu \right) \epsilon - Mc \int d\xi \frac{d}{d\xi} \left(\sqrt{1} \right) \epsilon$$

Q: why does second term vanish?

$$\frac{d}{d\xi} \left(\sqrt{1} \right) = \frac{1}{\sqrt{1}} \eta_{\mu\nu} \dot{X}^\mu \ddot{X}^\nu$$

~> Since now $\ddot{X}^\mu = 0$ which I haven't shown

$$\dot{X}^\mu \dot{X}_\mu > 0$$

$$P^2 = P_\mu P^\mu$$

$$P^2 = P_\mu P^\mu = M^2 c^2 \left(\frac{\eta_{\mu\nu} \dot{X}^\nu}{\sqrt{1}} \right) \left(\frac{\eta^{\mu\lambda} \dot{X}_\lambda}{\sqrt{1}} \right) = M^2 c^2$$

$$\sim \frac{dP^2}{d\xi} = 0 \Rightarrow M \text{ is invariant.}$$

$$P_\mu \frac{dP^\mu}{d\xi} = 0 = \left(\frac{\eta_{\mu\nu} \dot{X}^\nu}{\sqrt{1}} \right) \left(\frac{\ddot{X}^\mu}{\sqrt{1}} - \frac{\dot{X}^\mu \eta_{\sigma\lambda} \dot{X}^\sigma \ddot{X}^\lambda}{(\quad)^{3/2}} \right)$$

$$\eta_{\mu\nu} \dot{X}^\nu (\ddot{X}^\mu - (\dot{X}^\mu \eta_{\sigma\lambda} \dot{X}^\sigma \ddot{X}^\lambda) / \eta_{\mu\nu} \dot{X}^\nu \dot{X}^\nu)$$

$$= \eta_{\mu\nu} \dot{X}^\nu \ddot{X}^\mu - \eta_{\sigma\lambda} \dot{X}^\sigma \ddot{X}^\lambda = 0$$

$$\text{or } \dot{X}^\mu \ddot{X}_\mu = 0$$

Since $\dot{X}^\mu \neq 0$ and η is non-degenerate this implies

$$\ddot{X} = 0 \text{ as claimed.}$$

$$\ast \delta S = \int d\xi (\delta_x + \delta_\xi) (\quad)$$

$$= \int d\xi (Mc) \left(\frac{1}{\sqrt{1}} \right) (\eta_{\mu\nu} \dot{X}^\mu \delta \dot{X}^\nu) + \dots$$

$$= \int d\xi \frac{d}{d\xi} (P_\nu) \epsilon \dot{X}^\nu + \dots$$

$$\therefore \frac{\delta S}{\delta \dot{X}^\mu} = P_\mu$$

$$\boxed{P_\mu \dot{X}^\mu = 0} \quad (\text{Gauge identity})$$

$$\partial X^\mu \quad \hookrightarrow \quad p^\mu p_\mu = 0 = \frac{d}{d\tau}(p^\mu p_\mu)$$

$p^\mu p_\mu = M^2 c^2$: without using the eq. of motion, we $\hookrightarrow p^2 = \text{const}$
 See that $p^2 = \text{const.}$ is a consequence of rep. invariance

choices of ξ : τ - proper time

$$g_{\mu\nu} \dot{X}^\mu(\tau) \dot{X}^\nu(\tau) = 1$$

$$\Rightarrow S_{pp}[X(\tau)] = -Mc \int d\tau$$

Action of a point-particle is proportional to its worldline's

proper length
 $\hookrightarrow \frac{i}{\hbar} S = i \frac{Mc}{\hbar} \int d\tau = \frac{i}{\hbar c} \int d\tau$

$\int d\tau$: proper length is measured in units of reduced Comp. wave length.

(ii) Coordinate Time: $\xi = X^0 = cT$, gauge fixes $X^0(\xi) = \xi$
 $(1 - (\dot{X}^i)^2)$

(Physical/static gauge)

Can study the non-rel. limit

$$\dot{X}^i{}^2 = (\frac{v}{c})^2 \ll 1$$

$$\begin{aligned} S &= -Mc \int d(cT) \sqrt{1 - (\frac{v}{c})^2} \\ &= -Mc^2 \int dT \left(1 - \frac{1}{2}(\frac{v}{c})^2 + \mathcal{O}(\frac{v}{c}^4) \right) \\ &= \int dT \left(\frac{1}{2} M v^2 - M c^2 \right) \end{aligned}$$

$$S_{pp}[X^i(T)] = \int dT \left(\frac{1}{2} M v^2 - M c^2 \right)$$

* Massive point particle coupled to scalar gravity:

$$S = \frac{1}{c} \int d^4x \left\{ \frac{1}{2\alpha c^2} (\partial\phi)^2 + \frac{\phi}{c^2} T_{\text{matter}} + \mathcal{L}_{\text{matter}} \right\}$$

$$(T_{pp})_{\mu\nu} = \beta \frac{\delta S}{\delta g^{\mu\nu}}$$

$$(T_{\mu\nu})(x) = -Mc \int d\xi \frac{\dot{X}_\mu \dot{X}_\nu}{\sqrt{-1}} \delta^d[X(\xi) - x]$$

Note: $\text{Tr}(T_{\mu\nu}) = \mathcal{L} \dots$

E.M.-tensor couples the Lagrangian...

* In the static gauge $X^0 = cT = \xi$

$$T_{00} = -Mc \int c dT \left(\dot{c}^2 / \sqrt{-1} \right) \delta^d[X(\xi) - x]$$

$$= -M \int dT \left(\frac{1}{\sqrt{-1}} \right) \delta^d[X(\xi) - x]$$

$$= -M \left(\int dT \frac{1}{\sqrt{-1}} \right) \delta^d[X(\xi) - x]$$

$$= -Mc^2 \int d\tau \frac{1}{\sqrt{c^2 - v^2}} \delta^d[X(\xi) - x]$$

$$= -pc^2 \quad \text{as claimed earlier} \quad \left(\int d\tilde{t}(1) = 1? \right)$$

* Scalar Gravity: $\mathcal{L}_{\text{matter}} = S_{\text{pp}}$

$$\boxed{S[X^\mu(\xi), \bar{x}^\mu(\xi)] = \frac{1}{c^3} \int d^d x \frac{1}{2} (\partial\phi)^2 - Mc \int d\xi \left(1 + \frac{\phi(x)}{c^2} \right) \sqrt{\eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu}}$$

(Point particle coupled to Scalar Gravity)

* For low speeds in the static Gauge

$$(i) -Mc \int d\xi \left(1 + \frac{\phi}{c^2} \right) \left(1 - \frac{1}{2} \left(\frac{v}{c} \right)^2 \right)$$

$$= - \int d\xi \left(1 + \frac{\phi}{c^2} - \frac{1}{2} \left(\frac{v}{c} \right)^2 - \frac{1}{2} \frac{v^2 \phi}{c^4} \right) Mc^2$$

$$= \int d\xi \left(\frac{1}{2} M v^2 + \frac{1}{2c^2} M v^2 \phi - M \phi - Mc^2 \right)$$

* classically, in the non-rel. case: $\frac{1}{2c^2} M v^2 \phi$ is negligible $\quad v \rightarrow \infty$
 $\sim \int d\xi \left(\frac{1}{2} M v^2 - M \phi - Mc^2 \right)$

* Compares with ϕ being the Newtonian potential

$$\frac{\partial \mathcal{L}}{\partial \dot{\vec{x}}} = \vec{p}, \quad m\vec{v} = |\vec{p}| \quad \mathcal{L} = \vec{K} - \vec{T}$$

$$\vec{F} = -\nabla(-M\phi)$$

$$\vec{F} = M \nabla \phi$$

* Have relativistic corrections $\sim -\frac{1}{2} M v^2 \left(\frac{v^2}{c^2} \right) + \frac{1}{2} M v^2 \phi / c^2$

$$\leadsto \frac{1}{2} M v^2 \phi^2 / c^2 \quad F = \frac{\partial \mathcal{L}}{\partial x} = -M \left(\frac{v^2}{c^2} - 1 \right) \frac{\partial \phi}{\partial x}$$

$$\vec{F} = \left(1 - \frac{v^2}{c^2} \right) \nabla \phi$$

as $v \rightarrow c$ the force decreases and no longer feels a gravitational force compared to when $v=0$

The Action for a Massless point particle:

$$S[X^\mu(\xi), \gamma(\xi)] = -\frac{1}{2} Mc \int d\xi \sqrt{-\gamma} \left(\gamma^{-1} \eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu + 1 \right)$$

$$\xi \mapsto \xi'(\xi)$$

$\rightarrow$ Requires $\sqrt{-\gamma}$ to transform as a 1-form:

$$(\gamma'(\xi'))^{1/2} d\xi' = (\gamma(\xi))^{1/2} d\xi$$

$\rightarrow$ How is one suppose to see this? Obviously it's a volume form so it must be invariant but

$$S[X^\mu(\xi'), \dots] = -\frac{1}{2} Mc \int d\xi' \sqrt{\gamma(\xi')} \left(\gamma'^{-1}(\xi') (\dot{X}')^2 + 1 \right)$$

$$\delta S = 0 = -\frac{1}{2} Mc \int d\xi' \sqrt{\gamma(\xi')} \left(\gamma'^{-1}(\xi') (\dot{X}')^2 + 1 \right) -$$

$$\gamma'(\xi') \eta_{\mu\nu} \dot{X}^\mu(\xi') \dot{X}^\nu(\xi') = \gamma'(\xi') \eta_{\mu\nu} \dot{X}^\mu(\xi) \dot{X}^\nu(\xi)$$

$$\dot{X}^\mu(\xi') = \frac{\partial \xi}{\partial \xi'} \frac{\partial}{\partial \xi} (X^\mu(\xi) + \xi \dot{X}^\mu + \dots)$$

$$= \frac{\partial \xi}{\partial \xi'} \dot{X}^\mu$$

$$\therefore (\dot{X}^\mu)^2 = \left(\frac{\partial \xi}{\partial \xi'} \right)^2 \dot{X}^2 \Rightarrow \gamma'^{-1}(\xi') \left(\frac{\partial \xi}{\partial \xi'} \right)^2 = \gamma^{-1}(\xi)$$

$$\text{or } \gamma(\xi) = \left(\frac{\partial \xi'}{\partial \xi} \right)^2 \gamma'(\xi')$$

Similarly found from requiring

$$d\xi' \sqrt{\gamma'(\xi')} = d\xi \sqrt{\gamma(\xi)}$$

* Most S.T. books use $e(\xi) := \sqrt{\gamma(\xi)}$

$$\# \text{ E.O.M } \frac{\partial \mathcal{L}}{\partial \gamma} = \frac{1}{2} \frac{1}{\sqrt{\gamma}} (\gamma^{-1} \dot{X}^2 + 1) - \sqrt{\gamma} \gamma^{-2} \dot{X}^2 = 0$$

$$\frac{1}{2} \frac{1}{\sqrt{\gamma}} (\gamma^{-1} \dot{X}^2 + 1) + \sqrt{\gamma} (-1) \gamma^{-2} \dot{X}^2 = 0$$

$$\Rightarrow \frac{1}{2} (\gamma^{-1} \dot{X}^2 + 1) = \gamma^{-1} \dot{X}^2$$

$$\gamma^{-1} \dot{X}^2 = 1 \Rightarrow \boxed{\gamma = \eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu} \quad (\text{E.O.M})$$

$\Rightarrow$ Tells us that γ is simply the induced (pull-back) metric of the worldline

* Massless p.p. do not couple to gravity (as expected) also true in $d=4$ for $F^{\mu\nu} F_{\mu\nu}$ (Maxwell)

Self Coupled Scalar Gravity:

Q: Does gravity couple to all forms of matter/energy including grav. energy or only to non-grav. energies?

$\Rightarrow$ For Scalar SRTS grav does not couple to itself

* looking for a theory with $\partial^2 \phi = cT$

$$T = \text{Tr}(T_{\text{tot}})$$

T_{tot} : total E.M. tensor

Self coupled Scalar Gravity:

* In the free theory $\mathcal{T}(\phi)$ is quadratic in $\partial\phi$

$$\tilde{T}_{\mu\nu}|_{\text{free}} = \eta_{\mu\nu} (\partial\phi)^2 - \partial_\mu \phi \partial_\nu \phi$$

$$\tilde{S}S = \int d^4x \frac{1}{2} S (\partial\phi)^2 =$$

$$= \int d^4x \partial^\mu \phi \partial_\mu \phi = \int d^4x \partial_\mu (\partial^\mu \phi \phi) - \int d^4x \partial^2 \phi \phi$$

$$\sim \frac{1}{\sqrt{\gamma}} \delta S / \delta \gamma^{\mu\nu} = \frac{\delta}{\delta \gamma^{\mu\nu}} \left(\int d^4x \sqrt{\gamma} \gamma^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right)$$

$$= \frac{1}{2} \frac{1}{\sqrt{\gamma}} + \sqrt{\gamma} \partial_\mu \phi \partial_\nu \phi$$

$$\sim \text{def}(T) = \frac{1}{2} \text{tr}(\partial_\mu \partial^\mu \gamma) = |\gamma| \cdot \text{tr}(\gamma \delta \gamma)$$

$$\begin{aligned}
 &= |Y| Y^{\sigma\lambda} \delta Y_{\sigma\lambda} \\
 &\frac{1}{2} \sqrt{|g|} Y^{\sigma\lambda} Y_{\sigma\mu} Y_{\lambda\nu} (\partial\phi)^2 \\
 &= \frac{1}{2} \sqrt{|g|} Y^{\sigma\lambda} Y_{\sigma\lambda} (\partial\phi)^2 + \sqrt{|g|} \partial_\mu \phi \partial^\mu \phi \Big|_{Y=0} \\
 \therefore \boxed{(T_d)_{\mu\nu} = g_{\mu\nu} (\partial\phi)^2 - 2 \partial_\mu \phi \partial_\nu \phi}
 \end{aligned}$$

→ To obtain this same term in the E.O.M we must add $\phi(\partial\phi)^2$ to the Lagrangian.

However: this will contribute to the new $T_{\mu\nu}$ and to add this new term to the E.O.M we need $\phi^2(\partial\phi)^2$

which $\Rightarrow$ need an ∞ -th of corrections

Interaction terms $\sim \frac{\phi}{c^2} T_{\text{matter}}$

K : quadratic in first partial der.

$$V = V(\phi)$$

$$\mathcal{L}_{\text{matter}} = K - V$$

$$\text{Then from } T^\mu_\nu = \delta^\mu_\nu \mathcal{L} - \frac{2\mathcal{L}}{2\partial_\mu \phi} \partial^\mu \phi$$

$$T^\mu_\nu = \delta^\mu_\nu (K - V) - 2 \partial^\mu \phi \partial_\nu \phi$$

$$\text{Tr}(T^\mu_\nu) = T = d(K - V) - 2K$$

$$\therefore \boxed{T = (d-2)K - dV}$$

$$* S = \frac{1}{c} \int d^d x \left\{ \frac{1}{2\alpha c^2} (\partial\phi)^2 + K \left(1 + \frac{(d-2)\phi}{c^2} \right) - V \left(1 + \frac{d\phi}{c^2} \right) \right\}$$

• For the corrections mentioned above we expect

$$S = \frac{1}{c} \int d^d x \left\{ \frac{1}{2\alpha c^2} f(\phi) (\partial\phi)^2 + g(\phi) K - h(\phi) V \right\}$$

$$\text{Impose: } \partial^2 \phi = \alpha T$$

$$T = (d-2) \frac{1}{2\alpha c^2} f(\phi) (\partial\phi)^2 + (d-2) g(\phi) K - d h(\phi) V$$

E.O.M:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \frac{1}{\alpha c^3} f \partial^\mu \phi \quad \Bigg| \quad \frac{\partial \mathcal{L}}{\partial \phi} = \frac{1}{2\alpha c^3} f' (\partial\phi)^2 + \frac{g' K - h' V}{c}$$

$$\frac{1}{\alpha c^3} (f' \partial_\mu \phi \partial^\mu \phi + f \partial^2 \phi) = \frac{1}{2\alpha c^3} f' (\partial\phi)^2 + \frac{g' K - h' V}{c}$$

$$f \partial^2 \phi = \frac{1}{2} f' (\partial\phi)^2 + \alpha c^3 (g' K - h' V) - f' (\partial\phi)^2$$

$$\partial^2 \phi = \frac{\alpha c^2}{f} (g' K - h' V) - \frac{1}{2} \left(\frac{f'}{f} \right) (\partial \phi)^2$$

$$\partial^2 \phi = -\frac{1}{2} \left(\frac{f'}{f} \right) (\partial \phi)^2 + \frac{\alpha c^2}{f} g' K + \frac{\alpha c^2}{f} h' V$$

$$= \alpha T$$

$$= \alpha \left\{ (d-2) \frac{1}{2\alpha c^2} f (\partial \phi)^2 + (d-2) g K - d h V \right\}$$

$$(i) \frac{\alpha c^2}{f} g' = \alpha (d-2) g$$

$$(ii) -\alpha d h = -\frac{\alpha c^2}{f} h'$$

$$(iii) -\frac{1}{2} \left(\frac{f'}{f} \right) = \alpha (d-2) \frac{1}{2\alpha c^2} f$$

$$f' = -\frac{(d-2)}{c^2} f^2$$

$$\Rightarrow \int \frac{df}{f^2} = \int -\frac{(d-2)}{c^2} d\phi$$

$$\frac{1}{f} = \phi (d-2)/c^2 + a$$

$$\therefore f(\phi) = \frac{1}{a + \left(\frac{d-2}{c^2} \right) \phi}$$

$$g = f^{-1} b$$

$$h = (f^{-1} e)^{\frac{d}{d-2}}$$

$$a, b, e \in \mathbb{R}$$

$$\frac{\alpha c^2}{f} g' = \alpha (d-2) g$$

$$\frac{dg}{g} = \frac{(d-2)}{c^2} \left(a + \frac{(d-2)}{c^2} \phi \right) d\phi$$

$$\ln(g) = \frac{(d-2)}{c^2 a} \int \left(\frac{1}{1 + \beta \phi} \right) d\phi \quad \beta = \frac{d-2}{c^2 a}$$

$$= \frac{d-2}{c^2 a} \ln(1 + \beta \phi) \frac{1}{\beta} + \tilde{b}$$

$$g = (1 + \beta \phi) \tilde{b} = \frac{\tilde{b}}{a} \left(a + \frac{d-2}{c^2} \phi \right)$$

$$g(\phi) = \frac{\tilde{b}}{a} \left(a + \frac{d-2}{c^2} \phi \right) = b f^{-1}$$

$$g = b f^{-1}$$

$$h = (f^{-1} e)^{\frac{d}{d-2}}$$

$$a = b = e = 1 \text{ (weak field limit)}$$

$$S = \frac{1}{c} \int d^d x \left\{ \frac{1}{2\alpha c^2} \frac{(\partial \phi)^2}{\left(1 + \left[\frac{d-2}{c^2} \right] \phi \right)} + \left(1 + \frac{d-2}{c^2} \phi \right) K - \left(1 + \frac{d-2}{c^2} \phi \right)^{\frac{d}{d-2}} V \right\}$$

Gives rise to a new T

$\pi \rightarrow \pi + \alpha \phi$ $T \rightarrow T + \alpha \phi$ $H \rightarrow H + \alpha \phi$ $\rho \rightarrow \rho + \alpha \phi$ $\mu \rightarrow \mu + \alpha \phi$

The Geometric Einstein-Hilbert theory: based on a conformally flat

$$g_{\mu\nu} = \left[\frac{\Phi}{c^2} \right]^{\frac{2}{d-2}} \eta_{\mu\nu}$$

where $\Phi = c^2 \left[1 + \frac{d-2}{c^2} \phi \right]^{\frac{2}{d-2}}$ metric

- Dynamical conformal factor

$$S = \frac{1}{c} \int d^d x \left\{ \frac{2}{(d-2)^2 c^2} (\partial \Phi)^2 + \left(\frac{\Phi}{c^2} \right)^2 K - \left(\frac{\Phi}{c^2} \right)^{\frac{2d}{d-2}} V \right\}$$

Spin 2 field: imagine massless $\partial^2 h_{\mu\nu} = 0$

Doesn't have positive def energy unless we impose

$$\partial^\mu (h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h^\rho{}_\rho) = 0$$

- $h_{\mu\nu}$: symmetric $d(d+1)/2$

$\hookrightarrow \frac{d(d+1)}{2}$ contains spin-1 & spin-0 helicity states
 $\uparrow$ trace: $\eta^{\mu\nu} h_{\mu\nu}$

Massless states: $(E, 0, \dots, E)$

$\hookrightarrow$ Little group $SO(d-2)$

so it must be $(d-2)(d-1)/2 - \frac{2}{2} \leftarrow \text{trace}$
 $(d^2 - 3d + 2 - 2)/2 = d(d-3)/2$

* $\frac{d(d-3)}{2}$: total helicity states for a symmetric traceless (massless) tensor $h_{\mu\nu}$

$$d=4: 2 \rightarrow S_z \in \{-2, +2\}$$

$$\Rightarrow h^\mu{}_\mu = 0 \quad (\text{De Donder Gauge})$$

$$\partial^\mu h_{\mu\nu} \oplus h^\mu{}_\mu = 0 \quad \oplus \quad \partial^\mu (h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h^\rho{}_\rho) = 0$$

- Matter couples to gravity through T_{matter} in SRT and should expect:

$$\partial^2 h_{\mu\nu} = \chi (T_{\text{matter}})_{\mu\nu}$$

Note since T is conserved $\partial_\mu T^{\mu\nu} = 0$ then so too must the E.C.M: must be of the form

$$D^{\mu\nu}(h) = \chi T^{\mu\nu}$$

$$\rightarrow \partial_\mu D^{\mu\nu}(h) = 0 \quad (\text{off-shell})$$

$\hookrightarrow$ This theory must automatically satisfy the Bianchi/Gauge identity

when T_M is coupled to gravity only T_{tot} is conserved

- So we would expect to add T_{grav} to the RHS

$$\sim D^{\mu\nu}(h) = \chi (T_{\text{grav}} + T_M)$$

However, adding this means a cubic term must be added to $\mathcal{L}$ which in turn will add a correction to $T_{\mu\nu}$.
 $\Rightarrow \infty$ - # of corrections needed for a free spin 2 (Fierz-Pauli) theory.

Spin 1 SRFT: $(\partial^2 + m^2) A^\mu = 0$

Energy density is not positive definite unless we impose $\partial_\mu A^\mu = 0$

* Massive: $P^\mu = (E, 0, \dots, 0)$ irrep of $SO(d-1)$
 i.e. $(\frac{d-1}{2}, \frac{d-1}{2})$ of $SO(d-1)$ [$d=4$: $s = -1, 0, +1$]

Massless: $P^\mu = (E, 0, \dots, E)$ irrep of $SO(d-2)$
 $(\frac{d-2}{2}, \frac{d-2}{2})$ of $SO(d-2)$ [$d=4$: $s = -1, +1$]

$\leadsto$ Gauge condition fixes the gauge choice of "gauge" Θ :
 $A_\mu \rightarrow A_\mu + \partial_\mu \Theta$

Proof of Gauge fixing:

Note that E.O.M are invariant with $\partial^\mu A_\mu = 0$

$$(\partial^2 + m^2) A^\mu = 0 \rightarrow (\partial^2 + m^2)(A^\mu + \partial^\mu \Theta)$$

$$\Rightarrow \partial^\mu \partial^2 \Theta = 0$$

$$\text{Gauge: } \partial_\mu A^\mu = 0 \Rightarrow \partial_\mu (A^\mu + \partial^\mu \Theta) = \partial^2 \Theta = 0$$

$$\text{Note } K(\partial^2 \Theta) = |\vec{k}|^2 \Theta(k) = 0 \text{ and so } \Theta(k) = 0$$

Since it has no normalizable solⁿ in $\mathbb{R}^{D-1}$

$$\partial^2 A^\mu = j^\mu \iff \partial^2 \partial_\mu A^\mu = \partial_\mu j^\mu = 0$$

$\leadsto$ Want to construct a theory where the gauge condition occurs from the E.O.M:

$$D^\mu(A) + m^2 A^\mu = j^\mu$$

• Massless case: $\partial_\mu D^\mu(A) = 0$ (off-shell condition)

Solution is the Proca Lagrangian:

$$S[A] = \int d^d x \left[-\frac{1}{4} F^2 + \frac{m^2}{2} A^2 \right] \quad \left(\text{E.O.M} \right)$$

$$D^\mu(A) \equiv \partial_\mu F^{\mu\nu}, \quad F_{\mu\nu} = 2 \partial_{[\mu} A_{\nu]}$$

• $m=0$ case reduces to Maxwell

$$\partial_\mu D^\mu(A) = \partial_\mu \partial_\nu F^{\mu\nu} + m^2 \partial_\mu A^\mu = 0 \Rightarrow \partial_\mu A^\mu = 0$$

• Massless theory is invariant under $\delta A_\mu = \partial_\mu \lambda(x)$

$$\delta F^{\mu\nu} = \delta(\partial^\mu A^\nu - \partial^\nu A^\mu)$$

$$= \partial^\mu(\partial^\nu \Lambda) - \partial^\nu(\partial^\mu \Lambda) = 0$$

$$\Rightarrow \partial^\mu(A_\mu + \partial_\mu \Lambda) \stackrel{!}{=} 0$$

$$\partial^\mu \Lambda_\mu = -\partial^2 \Lambda$$

to fully fix this gauge we must introduce a new gauge cond.

* Maxwell Action & invariance: $S_M = \int d^d x (-\frac{1}{4} F^2)$

$$\begin{aligned} \delta S &= \int d^d x (-F^{\mu\nu} \partial_\mu \delta A_\nu) = \int d^d x \{ D^\mu(A) \delta A_\mu - \partial_\mu (F^{\mu\nu} \delta A_\nu) \} \\ &= \int d^d x \{ -\partial_\mu D^\mu(A) \Lambda - \partial_\mu (F^{\mu\nu} \partial_\nu \Lambda - D^\mu(A) \Lambda) \} \end{aligned}$$

* Argue: Λ & $\partial^{(n)} \Lambda$ vanish on the boundary

$$\text{Thus } \partial_\mu D^\mu = 0$$

• Consider more general parameters

$$\begin{aligned} (j_{N_1})^\mu &\equiv F^{\mu\nu} \partial_\nu \Lambda \\ (j_{N_2})^\mu &\equiv F^{\mu\nu} \partial_\nu \Lambda - D^\mu(A) \Lambda \end{aligned} \quad \left| \quad \begin{aligned} \partial_\mu j_{N_2}^\mu &= 0 \\ \text{(Noether currents)} \end{aligned} \right.$$

(j1): only conserved on-shell

(j2): cons. off-shell

⇒ On-shell - they are equivalent $D^\mu = 0$ (on-shell)

$$(j_2)^\mu = \partial_\nu (j_2^{\nu\mu}(A)) \quad j_2^{\mu\nu} \equiv F^{\mu\nu} \Lambda$$

i.e. it is exact

$$\begin{aligned} \text{* Conserved charge: } q(\Lambda) &\sim \int_{V_2} d^{d-1} \sum_\mu j^\mu(\Lambda) \\ &\sim \frac{1}{2} \int_{S_{d-2}} d^{d-2} \sum_{\mu\nu} (j_2)^{\mu\nu} = -\frac{1}{2} \int_{S_{d-2}} (d^{d-2} \sum_{\mu\nu}) F^{\mu\nu} \Lambda \end{aligned}$$

* For $\Lambda = 1$: q : is electric charge

$$q = \int_{\partial \Sigma} *F$$

$$S_j(A) = \int d^d x \left(-\frac{1}{4} F^2 - A_\mu j^\mu \right)$$

$$\int \delta(A_\mu j^\mu) = \int d^d x \partial_\mu \Lambda j^\mu = -\int d^d x (2 j^\mu \Lambda + \partial_\mu (j^\mu \Lambda))$$

or in full:

$$\delta S_j = \int d^d x \{ -\partial_\mu (D^\mu(A) - j^\mu) \Lambda - \partial_\mu \{ F^{\mu\nu} \partial_\nu \Lambda - [D^\mu(A) j^\mu] \} \Lambda \}$$

SREFT with a spin 2-particle:

* Start with a linear theory which contains the most general Lorentz invariant $\mathcal{L}$ which is quadratic in $\partial_\mu h^{\mu\nu}$

and impose the E.O.M.

ex. Spn-1: $S = \int d^d x \{ a \partial_\mu A_\nu \partial^\mu A^\nu + b \partial_\mu A_\nu \partial^\nu A^\mu \}$

impose

$$\partial_\mu D^\mu(A) = 0$$

$$\frac{\delta S}{\delta A^\mu} = 0$$

$$\begin{aligned} \frac{\delta S}{\delta A^\mu} &= a \left(\eta_{\nu\lambda} \delta_\mu^\sigma \delta_\rho^\lambda \partial^\mu A^\nu + \partial_\mu A_\nu \eta^{\mu\lambda} \delta_\lambda^\sigma \delta_\rho^\nu \right) \\ &\quad + b \left(\eta_{\nu\lambda} \delta_\rho^\lambda \delta_\mu^\sigma \partial^\nu A^\mu + \partial_\mu A_\nu \eta^{\mu\lambda} \delta_\lambda^\sigma \delta_\rho^\mu \right) \\ &= a \left(\partial^\sigma A_\rho + \partial^\sigma A_\rho \right) + b \left(\partial_\rho A^\sigma + \partial_\rho A^\sigma \right) \end{aligned}$$

$$\therefore \partial_\sigma \left(\frac{\delta S}{\delta A^\rho} \right) = a \left(\partial^2 A_\rho \right) + b \left(\partial_\rho \partial^\sigma A^\sigma \right)$$

Take $\partial^\rho ()$: $a \partial^2 \partial^\sigma A_\sigma + b \partial^2 \partial^\sigma A_\sigma = 0$

variables for $a = -b$

$$F^2 = (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu)$$

$$\begin{aligned} &= \partial_\mu A_\nu \partial^\nu A^\mu - \partial_\nu A_\mu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu + \partial_\nu A_\mu \partial^\mu A^\nu \\ &= 2 (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu) \end{aligned}$$

$$-\frac{1}{4} F^2 = -\frac{1}{2} ()$$

$\therefore a = -\frac{1}{2}$ yields Maxwell

* We now exactly this for $\phi_{\mu\nu}$ and impose $\partial_\mu D^\mu(h) = 0$
 $h_{\mu\nu}$: Symmetric + traceless

* Up to total derivative terms:

(a) $\partial_\mu h_{\nu\lambda} \partial^\mu h^{\nu\lambda}$

(d) $\partial_\rho h \partial^\rho h$

(c) $\partial_\rho h \partial^\rho h^{\rho\lambda}$

(b) $\partial_\rho h_{\mu\nu} \partial^\mu h^{\rho\nu}$

$$D = \eta^{\rho\sigma} \Lambda_\rho^\mu \eta_{\mu\nu} \Lambda_\sigma^\nu$$

$$* D = (\Lambda_\rho^\mu) (\Lambda_\mu^\sigma)$$

$$h \rightarrow \eta^{\mu\nu} h'_{\mu\nu} = \eta^{\mu\nu} \Lambda_\mu^\sigma \eta_{\sigma\lambda} \Lambda_\nu^\lambda$$

$$h' = \eta^{\sigma\lambda} h_{\sigma\lambda} = h$$

no These are the four possible gauge inv. combos

$$S = \int d^d x \{ a \partial^\rho h^{\mu\nu} \partial_\rho h_{\mu\nu} + b \partial^\mu h^{\nu\rho} \partial_\nu h_{\mu\rho} + c \partial^\mu h \partial^\lambda h_{\lambda\mu} + d \partial^\mu h \partial_\mu h \}$$

* E.O.M.: (aside)

$\int_M F \wedge * F$: Since $dF = 0$ (E.O.M. of F)

$d * F = 0$ (E.O.M. of $*F$)

Proof: Let $F^{(p+1)} = dA^{(p)} = * G^{(d-p-1)}$

$\int *G^{(d-p-1)} \wedge (C-1)^{p-1} G^{(d-p-1)}$
 E.O.M are $dG^{(d-p-1)} = 0$
 but $G^{(d-p-1)} = *F^{(p-1)}$ and so
 $d*F = 0$ as claimed

Now for S : $\frac{\delta \mathcal{L}}{\delta h_{\mu\nu}} = 0$

$$\begin{aligned}
 \frac{\delta \mathcal{L}}{\delta h_{bc}} &= 2a(\partial^a h_{bc}) + b(\partial_\mu h^\mu{}_\rho \delta^a{}_\mu \delta^{\rho c} + \delta^a{}_\mu \delta^{\mu c} \delta^{\rho c}) \\
 &\quad + c(\eta_{\rho\sigma} \delta^a{}_\mu \delta^{\rho c} \delta^{\sigma c} \partial^\lambda h^\mu{}_\lambda + \partial_\mu h \delta^a{}_\lambda \delta^{\mu c} \delta^{\lambda c}) \\
 &\quad + d(2\eta_{\lambda\sigma} \delta^a{}_\mu \delta^{\lambda c} \delta^{\sigma c} \partial^\mu h) \\
 &= 2a(\partial^a h_{bc}) + b(\partial_b h^a{}_c + \partial_c h^a{}_b) + c(\partial^\lambda h_\lambda{}^a \eta_{bc} + \partial_b h \delta^a{}_c) \\
 &\quad + 2d(\partial^a h \eta_{bc}) = D_{bc}^{(a)}(h)
 \end{aligned}$$

Now we need:

$$\begin{aligned}
 \partial_a \left(\frac{\delta \mathcal{L}}{\delta h_{bc}} \right) &= 2a(\partial^2 h_{bc}) + b(\partial_b \partial_a h^a{}_c + \partial_c \partial_a h^a{}_b) + c[\partial_a \partial^\lambda h_\lambda{}^a \eta_{bc} + (\partial_c \partial_b h)] + 2d(\partial^2 h \eta_{bc})
 \end{aligned}$$

$\therefore$ E.O.M:

$$D_{\mu\nu}(h) = 2a(\partial^2 h_{\mu\nu}) + b(\partial_\mu \partial_\sigma h^{\sigma\lambda} \eta_{\lambda\nu} + \partial_\nu \partial_\sigma h^{\sigma\lambda} \eta_{\lambda\mu}) + c[(\partial^\sigma \partial^\lambda h_{\lambda\sigma}) \eta_{\mu\nu} + \partial_\mu \partial_\nu h] + 2d(\partial^2 h \eta_{\mu\nu}) = 0$$

* Impose Divergentless identity ($\partial_\mu D^{\mu\nu} = \chi \partial_\mu T^{\mu\nu} \stackrel{!}{=} 0$) $a = \frac{1}{4}$

$$\begin{aligned}
 \partial_\mu D^{\mu\nu} &= \frac{1}{2}(\partial^2 \partial_\sigma h^{\sigma\nu}) + b(\partial^2 \partial_\sigma h^{\sigma\nu} + \partial_\mu \partial^\nu \partial_\sigma h^{\sigma\mu}) \\
 &\quad + c[\partial_\mu \partial^\sigma \partial^\lambda h_{\lambda\sigma} \eta^{\mu\nu} + \partial^2 \partial^\nu h] + 2d \partial^2 \partial_\mu h \eta^{\mu\nu} \stackrel{!}{=} 0 \\
 \Rightarrow (\frac{1}{2} + b)(\partial^2 \partial_\sigma h^{\sigma\nu}) + (b+c)(\partial^\nu \partial_\sigma \partial_\lambda h^{\sigma\lambda}) + (c+2d)(\partial^2 \partial^\nu h) &= 0
 \end{aligned}$$

$\therefore$
 $a = \frac{1}{4} \quad b = -\frac{1}{2} \Rightarrow c = \frac{1}{2} \Rightarrow d = -\frac{1}{4}$

more generally: $2a + b = 0$

$a: \quad b = -2a, \quad c = 2a, \quad d = -a$

Note for diff. ref.

E.O.M become: $(\partial^2 h_{\mu\nu}) - (\partial_\mu \partial_\sigma h^{\sigma\lambda} \eta_{\lambda\nu} + \partial_\nu \partial_\sigma h^{\sigma\lambda} \eta_{\lambda\mu})$

or

$$\boxed{D_{\mu\nu}(h) = \partial^2 h_{\mu\nu} - 2 \partial^\sigma \partial_{(\mu} h_{\nu)\sigma} + \partial_\mu \partial_\nu h - \eta_{\mu\nu} (\partial^2 h - \partial_\sigma \partial_\lambda h^{\sigma\lambda}) = 0}$$

* Can remove the trace without loss of information

$$D_p^p = \partial^2 h - \partial^\sigma (\partial^\mu h_{\mu\sigma} + \partial^\mu h_{\mu\sigma}) + \partial^2 h - D(\partial^2 h - \partial_\sigma \partial_\lambda h^{\sigma\lambda}) = 0$$

$$= 2\partial^2 h - 2 \partial^\sigma \partial^\mu h_{\mu\sigma} + D(\quad)$$

$$= (2-D) \partial^2 h + (D-2) \partial_\sigma \partial_\lambda h^{\sigma\lambda} = 0$$

$$= (2-D) (\partial^2 h - \partial_\sigma \partial_\lambda h^{\sigma\lambda}) = 0$$

$$D_p^p = (2-D) (\partial^2 h - \partial_\sigma \partial_\lambda h^{\sigma\lambda}) = 0$$

(i) $D=2$: makes this true

$$(ii) D \neq 2: \partial^2 h - \partial_\sigma \partial_\lambda h^{\sigma\lambda} = 0$$

$$\partial_\sigma (\partial^\sigma h - \partial_\lambda h^{\sigma\lambda}) = 0$$

$$\partial^\sigma h - \partial_\lambda h^{\sigma\lambda} = (\text{const})$$

$$\tilde{D}_{\mu\nu}(h) \equiv D_{\mu\nu}(h) + \frac{1}{2-D} \eta_{\mu\nu} D^\mu{}_\mu = \partial^2 h_{\mu\nu} + \partial_\mu \partial_\nu h - 2 \partial^\lambda \partial_{(\mu} h_{\nu)\lambda} = 0$$

* Can also introduce the term

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$$

$$\text{s.t. } D^{\mu\nu}(\tilde{h}) = 0$$

$$\Rightarrow \text{Similar to spin 1: } D^\mu(A) = \partial_\nu F^{\nu\mu}$$

We want to write D^μ as the divergence of some

Ω :

$$D^{\mu\nu}(\tilde{h}) = 2 \partial_\mu \Omega^{\nu\mu}$$

$$\Omega_T^{\nu\mu} = \Omega_T^{[\nu\mu]} = -\partial_\sigma H^{\mu\sigma\nu\rho}$$

$$\Omega_L^{\nu\mu} = \Omega_L^{[\nu\mu]} = -\partial_\sigma K^{\nu\sigma\mu\rho}$$

$$\Omega_{AD}^{\nu\mu} = \Omega_{AD}^{[\nu\mu]} = -\partial_\sigma K^{\mu\nu\rho\sigma}$$

$$\bullet K^{\mu\nu\rho\sigma} \equiv \frac{1}{2} (\eta^{\mu\sigma} \tilde{h}^{\nu\rho} + \eta^{\nu\rho} \tilde{h}^{\mu\sigma} - \eta^{\mu\rho} \tilde{h}^{\nu\sigma} - \eta^{\nu\sigma} \tilde{h}^{\mu\rho})$$

$$H^{\mu\sigma\nu\rho} \equiv \frac{1}{2} (\eta^{\sigma\rho} \tilde{h}^{\mu\nu} + \eta^{\mu\nu} \tilde{h}^{\sigma\rho} - \eta^{\mu\rho} \tilde{h}^{\nu\sigma} - \eta^{\nu\sigma} \tilde{h}^{\mu\rho})$$

$$K^{\mu\nu\rho\sigma} = -K^{\mu\nu\sigma\rho}$$

$$H^{\mu\nu\rho\sigma} = H^{\mu\nu\sigma\rho}$$

* K has the same symmetries as the Riemann tensor

$$K^{\mu\nu\rho\sigma} = K^{\rho\sigma\mu\nu}, \quad K^{\mu\nu\rho\sigma} = K^{\nu\mu\sigma\rho}, \quad K^{\mu\nu\rho\sigma} = K^{\rho\sigma\mu\nu}$$

$$\bullet \text{ Note } \Omega_T^{\nu\mu} = \frac{\partial \mathcal{L}_{FP}}{\partial (\partial_\mu h_{\nu\rho})}$$

Gauge Symmetry of FP action:

$$\delta S_{FP} = -\frac{1}{2} \int d^d x \, D^{\mu\nu} \delta h_{\mu\nu}$$

→ This is a diffeomorphism invariance under ϵ_μ

$$x'^\mu = x^\mu - \epsilon^\mu$$

$$(\mathcal{L}_\epsilon g)_{\mu\nu} = g'_{\mu\nu}(x) - g_{\mu\nu}(x)$$

• Note

$$g'_{\mu\nu}(x') = \frac{\partial x^\sigma}{\partial x'^\mu} \frac{\partial x^\lambda}{\partial x'^\nu} g_{\sigma\lambda}(x)$$

$$\begin{aligned} \bullet \quad g_{\mu\nu}(x) &= \partial_\mu (x^\sigma - \epsilon^\sigma) \partial_\nu (x^\lambda - \epsilon^\lambda) g_{\sigma\lambda}(x') \\ &= (\delta_\mu^\sigma - \partial_\mu \epsilon^\sigma) (\delta_\nu^\lambda - \partial_\nu \epsilon^\lambda) g_{\sigma\lambda}(x') \\ &= (\delta_\mu^\sigma \delta_\nu^\lambda - \partial_\mu \epsilon^\sigma \delta_\nu^\lambda - \partial_\nu \epsilon^\lambda \delta_\mu^\sigma) g_{\sigma\lambda}(x') \\ &= (\delta^\sigma_\nu + \dots) (g'_{\sigma\lambda}(x) - \epsilon^\mu \partial_\mu g'_{\sigma\lambda}(x) - \mathcal{O}(\epsilon^2)) \end{aligned}$$

$$\begin{aligned} &= g'_{\mu\nu}(x) - \epsilon^\sigma \partial_\sigma g'_{\mu\nu}(x) - \partial_\mu \epsilon^\sigma \delta_\nu^\lambda g'_{\sigma\lambda}(x) - \partial_\nu \epsilon^\lambda \delta_\mu^\sigma g'_{\sigma\lambda}(x) \\ &= g'_{\mu\nu}(x) - \epsilon^\sigma \partial_\sigma g'_{\mu\nu}(x) - \partial_\mu \epsilon^\sigma g'_{\sigma\nu}(x) - \partial_\nu \epsilon^\lambda g'_{\mu\lambda}(x) \\ &= g'_{\mu\nu}(x) - \epsilon^\sigma \partial_\sigma g'_{\mu\nu} - 2 \partial_{[\mu} \epsilon_{\nu]} \end{aligned}$$

$$\tilde{\delta} = \delta + \tilde{\delta} X^\mu \partial_\mu$$

The above really should be

$$\delta h_{\mu\nu} = h'_{\mu\nu}(x') - h_{\mu\nu}(x) = 2 \partial_{[\mu} \epsilon_{\nu]}$$

* Action invariant under $\delta h_{\mu\nu} = -2 \partial_{[\mu} \epsilon_{\nu]}$

• Need ϵ_μ to remove 2d of the D.O.F. of h
Since we have not yet assumed traceless and require $h=0$ as a consistency condition.

• Traceless (transverse) gauge: $\partial_\mu h^{\mu\nu} = h = 0$
which leads to the E.O.M:

$$D_{\mu\nu}(x)_{TT} = \partial^2 h_{\mu\nu} = 0$$

• De-Donder (Harmonic) gauge: $\partial_\mu \bar{h}^{\mu\nu} = \partial_\mu (h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h) = 0$
E.O.M:

$$D_{\mu\nu}(x)_{DD} = \partial^2 \bar{h}_{\mu\nu} = 0$$

→ Transverse $\Rightarrow$ De-Donder (but not conversely)

inverse also doesn't fully fix the gauge as ϵ^μ is not completely determined: ϵ can be $\epsilon^\mu = \partial^\mu \tilde{\epsilon}$ or $\partial^2 \tilde{\epsilon} = 0$

* Want to find a massive theory:

$$(\partial^2 + m^2) h^{\mu\nu} = 0$$

$$h = 0 \quad \leftarrow \text{I constant}$$

$$\partial^\mu h_{\mu\nu} - \frac{1}{2} \partial_\nu h = 0 \quad \leftarrow \text{D-constants on } \mu$$

$\hookrightarrow (D+1)$ -constraints on $h_{\mu\nu}$

* $h_{\mu\nu}$: symmetric $(d^2 - d)/2 + d = \frac{d^2 - d + 2d}{2} = \frac{d(d+1)}{2}$

$$\therefore (d^2 + d - 2d - 2)/2 = (d^2 - d - 2)/2 = (d-2)(d+1)/2$$

* For $d=4$: $(D.O.F) = 5 \quad S_2 = -2, -1, 0, 1, 2$

* The Bianchi/Gauge identity: $\partial_\mu D^{\mu\nu}(h) = 0$ implies we can add terms which vanish up to our constraints

i.e. $\partial_\mu D^{\mu\nu}(h) = 0$ implies

$$\tilde{D}^{\mu\nu}(h) = D^{\mu\nu} + a(h^{\mu\nu}) + b\eta^{\mu\nu}h = 0$$

$$\partial_\mu \tilde{D}^{\mu\nu}(h) = \partial_\mu D^{\mu\nu} + a \partial_\mu h^{\mu\nu} + b \partial^\nu h = 0$$

$$\therefore a \partial_\mu h^{\mu\nu} + b \partial^\nu h = 0$$

$$\leadsto \text{Gauge has } \partial_\mu h^{\mu\nu} = h = 0$$

Note however

$$\eta^{\mu\nu} \tilde{D}_{\mu\nu} = D + a(h) + b d h = 0$$

$$D + h(a + bd) = 0$$

$\leadsto$ Not too sure of the argument... I will note that the original E.D.M have

$$- \eta_{\mu\nu} (\partial^2 h - \partial_\lambda \partial_\sigma h^{\lambda\sigma})$$

which, being proportional to η should tell us of its "non-importance" regarding the theory. In fact the R.T. is

$$R_{\mu\nu} = D_{\mu\nu} - \text{"this term"} \text{ for massless linearized gravity}$$

$$- \partial^2 h + \partial_\lambda \partial_\sigma h^{\lambda\sigma} = \partial^\alpha (\partial^\lambda h_{\lambda\sigma} - \partial_\sigma h)$$

$$= \partial^\sigma (\partial^\lambda h_{\lambda\sigma} - \partial^\lambda \eta_{\lambda\sigma} h) = \partial^\sigma \partial^\lambda (h_{\lambda\sigma} - \eta_{\lambda\sigma} h)$$

* And the remaining term is what is we are adding us a mass-term:

$$\sim m^2 (h_{\mu\nu} - \eta_{\mu\nu} h)$$

Note also this extra term is exactly $\frac{1}{2}$ trace of D

$$D = D^\mu_\mu = (2-d)(\partial^2 h - \partial_\mu \partial_\nu h^{\mu\nu})$$

$$\Rightarrow (2-d)(\partial^2 h - \partial_\mu \partial_\nu h^{\mu\nu}) + m^2 h = 0$$

$$(\partial^2 - \frac{m^2}{(d-2)}) h - \partial_\mu \partial_\nu h^{\mu\nu} = 0$$

$$\bullet \text{ If } \partial^\mu h_{\mu\nu} = \partial_\nu h \Rightarrow \partial^\mu (h_{\mu\nu} - \eta_{\mu\nu} h) = 0$$

... Not too sure ...

Massive F.P.:

$$S = \int d^d x \left\{ \frac{1}{4} \partial^\mu h^{\mu\nu} \partial_\mu h_{\nu\lambda} - \frac{1}{2} \partial^\mu h^{\mu\nu} \partial_\mu h_{\nu\lambda} + \frac{1}{2} \partial^\mu h \partial_\mu h - \frac{1}{4} \partial^\mu h \partial_\mu h - \frac{1}{4} m^2 (h^{\mu\nu} h_{\mu\nu} - h^2) \right\}$$

$$D^{\mu\nu}_{(m)}(h) = D^{\mu\nu}(h) + m^2 (h^{\mu\nu} - \eta^{\mu\nu} h) = 0$$

mass term

$$\text{On-shell/Gauge: } h = 0, \partial_\mu h^{\mu\nu} = 0, (\partial^2 + m^2) h_{\mu\nu} = 0$$

$$\bullet \text{ Mass-term } h^{\mu\nu} (h_{\mu\nu} - \eta_{\mu\nu} h) \quad [2, 0, \dots, 0]$$

* Can use the Gauge symm. to derive conserved currents

$$\delta S_{FP} = \int d^d x \left(\frac{\delta S}{\delta h^{\mu\nu}} \delta h^{\mu\nu} + \frac{\delta S}{\delta \partial_\mu h^{\mu\nu}} \delta (\partial_\mu h^{\mu\nu}) \right)$$

$$= \int d^d x \left\{ \frac{\delta S}{\delta h^{\mu\nu}} \delta h^{\mu\nu} - \partial_\sigma \left(\frac{\delta S}{\delta \partial_\sigma h^{\mu\nu}} \right) \delta h^{\mu\nu} + \partial_\sigma \left(\frac{\delta S}{\delta \partial_\sigma \partial_\mu h^{\mu\nu}} \right) \delta h^{\mu\nu} \right\}$$

$$S_{FP}: \text{Massless theory thus } \frac{\delta S}{\delta h^{\mu\nu}} = 0 \quad \left| \begin{array}{l} \Omega_T^{\mu\nu\rho} = \Omega_T^{\rho\mu\nu} = \\ = -2 H^{\rho\sigma\mu\nu} \end{array} \right.$$

$$\delta S_{FP} = \int d^d x \left\{ \partial_\sigma \left(\frac{\delta S}{\delta \partial_\sigma h^{\mu\nu}} \right) \delta h^{\mu\nu} - \partial_\sigma \left(\frac{\delta S}{\delta \partial_\sigma \partial_\mu h^{\mu\nu}} \right) \delta h^{\mu\nu} \right\}$$

$$\delta h_{\mu\nu} = -2 \partial_{[\mu} \epsilon_{\nu]}, \quad -\frac{1}{2} \partial_\sigma \left(\frac{\delta S}{\delta \partial_\sigma h^{\mu\nu}} \right) = D_{\mu\nu}$$

$$\Rightarrow \int d^d x \left\{ \partial_\sigma \left[\Omega_T^{\mu\nu\rho} (-2 \partial_{[\mu} \epsilon_{\nu]} \right] + 2 D_{\mu\nu}(h) \partial^{\mu\nu} \epsilon^\nu \right\}$$

$$= \int d^d x \left\{ \partial_\sigma \left[2 \partial_\rho H^{\rho\sigma\mu\nu} \partial_\mu \epsilon_\nu \right] - D^{\sigma\nu}(h) \partial_\sigma \epsilon_\nu \right\}$$

$$= \int d^d x \left\{ \partial_\sigma \left[2 \partial_\rho H^{\rho\sigma\mu\nu} \partial_\mu \epsilon_\nu - D^{\sigma\nu} \epsilon_\nu \right] + \partial_\sigma D^{\sigma\nu} \epsilon_\nu \right\}$$

Note:

$$\textcircled{a} \partial_\sigma \partial_\rho H^{\rho\sigma\mu\nu} \partial_\mu \epsilon_\nu = \partial_\sigma \partial_\rho H^{\rho\sigma\mu\nu} \partial_\mu \epsilon_\nu + \partial_\rho H^{\rho\sigma\mu\nu} \partial_\sigma \partial_\mu \epsilon_\nu$$

$$\partial_\mu \epsilon^\mu = \partial_\sigma \partial_\rho H^{\rho\sigma\mu\nu} \partial_\mu \epsilon_\nu + \partial_\rho H^{\rho\sigma\mu\nu} \partial_\sigma \partial_\mu \epsilon_\nu - \partial_\sigma H^{\rho\sigma\mu\nu} \partial_\rho \partial_\mu \epsilon_\nu - H^{\rho\sigma\mu\nu} \partial_\sigma \partial_\rho \partial_\mu \epsilon_\nu$$

Take comp and int. by parts in $\textcircled{a}$

$$D^{\mu\nu} = 2 \partial_\sigma \partial_\rho H^{\rho\sigma\mu\nu}$$

$$j_{N_2}^\mu(\epsilon) = j_{N_1}^\mu(\epsilon) + D^{\mu\nu}(h) \epsilon_\nu$$

$$j_{N_1}^\mu(\epsilon) = 2 \partial_\sigma H^{\mu\sigma\rho\nu} \partial_\nu \epsilon_\rho - 2 H^{\mu\sigma\rho\nu} \partial_\sigma \partial_\nu \epsilon_\rho$$

$$\bullet j_{N_2}^\mu(\epsilon) = 2 j_{N_1}^{\mu\nu}(\epsilon)$$

$$j_{N_2}^{\mu\nu}(\epsilon) = -2 \partial_\sigma K^{\mu\nu\rho\sigma} \epsilon_\rho + 2 (H^{\mu\nu\rho\sigma} + K^{\mu\nu\rho\sigma}) \partial_\sigma \epsilon_\rho$$

• Use to calculate conserved charges ass. with ϵ_μ

Coupling to matter: Comes from $D^{\mu\nu}(h) = \chi T^{\mu\nu}_{\text{matter}}$

$$\mathcal{L} = \mathcal{L}_{\text{FP}} + \mathcal{L}_{\text{matter}}(\varphi, h)$$

where $\mathcal{L}_{\text{matter}}(\varphi, h) = \mathcal{L}_{\text{matter}}(\varphi) + \frac{1}{2} \chi h_{\mu\nu} T^{\mu\nu}(\varphi)$

$\rightarrow$ E.O.M. of $h_{\mu\nu}$ then follows from $\mathcal{L}$

$\rightarrow$ Must have conserved $T^{\mu\nu}$ (Gauge identity) $\partial_\mu T^{\mu\nu}(\varphi) = 0$

• Diffeomorphism invariant: $\delta h_{\mu\nu} = -2 \partial_{(\mu} \xi_{\nu)}$ iff $\partial_\mu T^{\mu\nu} = 0$

Questions:

(i) Which $T^{\mu\nu}$ is to be used (i.e. field type?)

(ii) Does $\partial_\mu T^{\mu\nu}$ stay consistent with modifications to φ E.O.M

(i) The Belinfante E.M. tensor is generally considered as the E.M. tensor which gravity couples to minimally

* Here the Belinfante E.M. tensor is the same as that found via the Rosenfeld prescription $T_{\mu\nu} = \frac{-2}{18} \frac{\delta \mathcal{S}}{\delta \gamma^{\mu\nu}}|_{\gamma=0}$ and since the matter sector looks like

$$S_m + \int d^4x \chi h_{\mu\nu} \frac{\delta \mathcal{S}[\varphi, \chi]}{\delta \gamma^{\mu\nu}}|_{\gamma=0}$$

it appears as if this is the first steps in a series expansion around $\{\gamma_{\mu\nu} = \eta_{\mu\nu} + \chi h_{\mu\nu}\}$

$$S_m[\varphi, \gamma_{\mu\nu}] = S_m[\varphi, \eta] + \int d^4x \chi h_{\mu\nu} \left. \frac{\delta \mathcal{S}}{\delta \gamma^{\mu\nu}} \right|_{\gamma=0} + \int d^4x d^4x' h_{\mu\nu}(x) h_{\rho\sigma}(x') \left. \frac{\delta^2 \mathcal{S}}{\delta \gamma^{\mu\nu} \delta \gamma^{\rho\sigma}} \right|_{\gamma=0} + \dots$$

Grav field of a massive point particle: $T^{\mu\nu} = -M c \int d\tilde{x} \frac{\dot{x}^\mu \dot{x}^\nu}{m_{\text{eff}} \sqrt{-\dot{x}^2}} \delta^{(4)}(\tilde{x}(\tilde{\xi}) - x)$

$$\begin{aligned} S_{\text{pp}}[\tilde{x}^\mu(\tilde{\xi})] &= -M c \int d\tilde{\xi} \sqrt{-\dot{\tilde{x}}^2} + \frac{1}{c} \int d\tilde{\xi} \frac{1}{2} \chi T^{\mu\nu} h_{\mu\nu} \\ &= -M c \int d\tilde{\xi} \left(\sqrt{-\dot{\tilde{x}}^2} + \frac{\dot{\tilde{x}}^\mu \dot{\tilde{x}}^\nu h_{\mu\nu}}{\sqrt{-\dot{\tilde{x}}^2}} \right) \\ &= -M c \int d\tilde{\xi} \left(m_{\mu\nu} + \frac{1}{2} \chi h_{\mu\nu} \right) \frac{\dot{\tilde{x}}^\mu \dot{\tilde{x}}^\nu}{\sqrt{m_{\mu\nu} \dot{\tilde{x}}^\mu \dot{\tilde{x}}^\nu}} \end{aligned}$$

* Invariant under gauge transformations only if $\partial_\mu T^{\mu\nu} = 0$

• Is invariant to lowest order in coupling χ without this assumption. if we therefore treat the particle as a point particle, we can do

$$\delta_\epsilon X^\mu = X^\mu(X) \quad (\text{infinitesimal GCT})$$

• First relation between gauge symmetry of the F.P. field and spacetime transformations

• There are fields $h_{\mu\nu}$ that are gauge equivalent to zero

ex: $h_{\mu\nu} = b_{\mu\nu} + a_{\mu\nu\rho} X^\rho$

cancelled by $\epsilon_\mu = \frac{1}{2}(b_{\mu\nu} X^\nu + a_{\mu\nu\rho} X^\nu X^\rho)$

* By a change of coordinates: we can remove certain grav. fields

$$D^{\mu\nu}(h) = (X/c) T^{\mu\nu}$$

Take the time to write

